S07 B IMPACT NOISE MANAGEMENT | O (MAX : 3 PT)

Intent : Reduce the level of impact noise radiation by designing resilient floors.

Summary : This WELL feature requires projects to manage background noise levels by demonstrating compliance with impact noise mitigation techniques.

Issue : Sound can transmit between rooms within a building as structure-borne impact noise. This impact noise travels through structures (e.g., walls, floors, columns, piping) as vibrations that are then radiated as airborne noise to a listener.[1] Impact noise typically derives from objects impacting a floor (e.g., footfall, machinery, gym equipment) and can result in workplace distractions, sleep disturbance or disrupted focus.[2,3]

Solutions : The overall construction of a building influences impact noise radiation levels. For example, a building that utilizes a light-weight floor construction (e.g., wood truss, cross-laminated timber, steel frame) generally exhibits higher degrees of impact noise radiation between floors.[4–7] Conversely, buildings constructed with resilient, composite floor-ceiling construction (e.g., thick concrete slab, suspended ceiling, floor with an underlayment) generally exhibit lower degrees of impact noise radiation. The performance of floor-ceiling materials can be measured using the following metrics: Impact Insulation Class Rating (IIC), Normalized Impact Sound Rating (NISR) or Weighted Standardized Impact Sound Pressure Level (L_{nTw}).

PART 1 SPECIFY IMPACT NOISE REDUCING FLOORING (MAX : 1 PT)

For All Spaces:

The following requirements are met:

a. For the following room types within the project boundary the floor-ceiling construction achieves the following minimum Impact Insulation Class (IIC) ratings with materials tested in accordance with ASTM E492-09, ISO 717.2 or predicted results determined through use of sound transmission modeling software (L_{nTw} may be used as an equivalent metric and values may be determined by subtracting the IIC values listed below from 110):

Room Type	Location of Applicable Floor-Ceiling Assembly	Minimum Impact Insulation Class (IIC)[103]
Quiet zones (except areas for concentration)	Above	55
Areas for fitness	Below	50
Enclosed areas for concentration and conferencing	Above	50
Open areas for concentration	Above	45
Areas for retail and dining	Below	45

Note :

This requirement does not apply to floor/ceiling assemblies that separate a relevant room type from a parking garage or non-occupiable space (e.g., a quiet zone that is vertically adjacent to a roof, equipment room or attic).

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

PART 2 MEET THRESHOLDS FOR IMPACT NOISE RATING (MAX : 2 PT)

For All Spaces:

The following requirements are met:

a. For the following room types within the project boundary the floor-ceiling construction achieves the following Normalized Impact Sound Ratings (NISR), as measured by a professional in acoustics, in accordance with ASTM E1007-19, ISO 16283 or equivalent (L_{nTw} may be used as an equivalent metric and may be determined by subtracting the NISR values listed below from 110):

Room Type	Location of Applicable Floor-Ceiling Assembly	Tier 1 1 point Minimum NISR[1]	Tier 2 2 points Minimum NISR[1]
Quiet zones (except areas for concentration)	Above	52	57
Areas for Fitness (If space is within the project boundary)	Below	47	52
Enclosed Areas for Concentration and Conferencing	Above	47	52

Open Areas for Concentration	Above	42	47
Areas for Retail and Dining (If space is within the project boundary)	Below	42	47

Note :
This requirement does not apply to floor/ceiling assemblies that separate a relevant room type from a non-occupiable space (e.g., a quiet zone that is vertically adjacent to a roof, equipment room or attic).

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

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